

Program : Diploma in Cloud Computing And Big Data	
Course Code : 4276	Course Title: Network and Operating System Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To introduce network related commands and Demonstrate traffic analysis using tools such as Wireshark .
- To Configure a simple router.
- To implement interprocess communication and process synchronization concepts.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Inter process communication, Process synchronization		Operating Systems	S3
Linux installation and basic linux command		Linux System Administration Lab	S3

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate hardware components in network and it's configuration	8	Applying
CO2	Experiment with basic networking commands and demonstrate Traffic Analysis using network monitoring tools.such as Wireshark.	6	Applying
CO3	Illustrate basic Network and DHCP server configuration using a router.	12	Applying

CO4	Implement interprocess communication and process synchronization concepts	13	Applying
	Lab Test	06	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2			2			
CO2	2	2		3			
CO3	2		2	2			
CO4	2		3			2	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Demonstrate hardware components in network and it's configuration		
M1.01	Relate various types of networking cables and connectors	2	Applying
M1.02	Choose various networking devices such as switch, router, wireless router etc.	1	Applying
M1.03	Demonstrate Crimping	2	Applying
M2.02	Experiment a LAN connection using static IP address	3	Applying
CO2	Experiment with networking commands and demonstrate Traffic Analysis using network monitoring tools such as Wireshark.		

M2.01	Experiment with basic networking commands such as ping, ifconfig, traceroute, netsat etc	2	Applying
M2.02	Use wireshark to analysis network traffic	4	Applying
	Lab Test – I	3	
CO3	Illustrate basic Network and DHCP server configuration using a router.		
M3.01	Demonstrate to configure the DHCP settings in a simple router, IP range , subnet configuration, IP reservation	2	Applying
M3.02	Illustrate Wifi configuration , Mac binding, different authentication methods	2	Applying
M3.03	Use static routing tables in routers using network simulation tools	3	Applying
M3.04	Complete the Installation and configuration of following services in a linux machine such as SSH,FTP server ,DHCP server and DNS server ,Web server	5	Applying
C04	Implement inter Process Communication and Process Synchronization concepts		
M4.01	Experiment basic socket programming using python and also establish a simple client server communication using TCP and UDP protocols	4	Applying
M4.02	Use pipes for Interprocess communication	3	Applying
M4.03	Illustrate semaphore for process synchronization	3	Applying
M4.04	Discover Open Ended Experiments**	3	Applying
	Lab Test – II	3	

Sample open Ended Projects

1. Setup a LAN for real world situation
2. Using Wireshark observe Three Way Handshaking Connection Establishment, Data Transfer and Three Way Handshaking Connection Termination in client server communication using TCP

Text / Reference

T/R	Book Title/Author
T1	T1 Vikas Gupta, Comdex Hardware and Networking Course Kit, Dreamtech Press, 2018

Online Resources

Sl No	Website Link
1	https://www.wireshark.org/docs/wsug_html_chunked/ChapterWork.html